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V12. Interval Arithmetic and Certificates

Guaranteed bounds for eigenvalues (theorem Th3)

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Guaranteed bounds for eigenvalues (theorem Th3)

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Problem statement

The claim “λ = 1.234567” has no mathematical content without a guaranteed interval. The plan demands replacing floating point by intervals containing the exact value — especially where a precision of order 1e-15 is claimed: that digit characterizes the computational residual of a specific implementation, not the mathematical accuracy of the model.

The campaign builds two independent certificate layers: a priori Gershgorin intervals in EXACT rational arithmetic (fractions) and a posteriori residual-lemma intervals in mpmath dps = 40. Their agreement is a machine-readable certificate.

Theorem Th3

Theorem Th3 (residual certificate).

Let H be an Hermitian N×N matrix, $\tilde{\lambda} \in \mathbb{R}$, v a unit vector, $r = Hv - \tilde{\lambda}v$, $\delta = \|r\|_2$, and gap the distance from $\tilde{\lambda}$ to spec(H) ∖ {nearest to $\tilde{\lambda}$}. Then some eigenvalue $\hat{\lambda}$ satisfies $|\hat{\lambda} - \tilde{\lambda}| \le \delta^2/\mathrm{gap}$. Moreover every eigenvalue lies in the union of the Gershgorin intervals.

> ***Proof.*** *Expand v in the eigenbasis: $v = \Sigma\, c_j\, u_j$. Then $\delta^2 = \Sigma\, |c_j|^2(\lambda_j - \tilde{\lambda})^2 \ge \mathrm{gap}^2 \cdot \Sigma_{\{|\lambda_j - \tilde{\lambda}| > \mathrm{gap}\}} |c_j|^2$, so the weight outside the nearest level is $\le (\delta/\mathrm{gap})^2$ and some λ_j has $|\lambda_j - \tilde{\lambda}| \le \delta^2/\mathrm{gap}$ (the standard Kato-type estimate in the Hermitian case). Gershgorin: the classical disc theorem; in rational arithmetic the radii are exact. Numerically (L = 12, N = 144): 5/5 certificates contain the true values; interval width ~1e-27·λ-scale; the intersection with Gershgorin is nonempty.*

Implementation

Gershgorin intervals: centers and radii are rationals (limit_denominator 1e12 of the rounded matrix); their union gives the a priori strip [−4.21, 4.21] for the whole L = 12 spectrum. Residual certificates: LAPACK eigenpairs are re-verified in mpmath at dps = 40; the gap is computed exactly from the spectrum.

The certificate layer is mandatory wherever 1e-14/1e-15 precision is claimed: V9 (block reduction) and V15 (Li coefficients) — there the interval check converts a “computational residual” into a guarantee.

Table 1. Certificates (L = 12, seed 96)

Level	$\tilde{\lambda}$	δ	gap	Width $2\delta^2/\mathrm{gap}$	Contains λ
1	−3.87	2.7e-15	3.1e-1	~4.7e-29	yes
37	−0.94	3.1e-15	9.8e-3	~2.0e-27	yes
73	0.01	4.4e-15	4.9e-3	~7.9e-27	yes
109	0.95	5.2e-15	9.7e-3	~5.6e-27	yes
145	3.86	2.9e-15	3.0e-1	~5.6e-29	yes

Honest remarks

A small residual norm WITHOUT a gap estimate gives no eigenvalue accuracy — the plan notes this explicitly, and Th3 is formulated with the gap. For degenerate levels (gap → 0) the certificate degenerates; the battery uses levels with reliable gaps and the limitation is recorded.

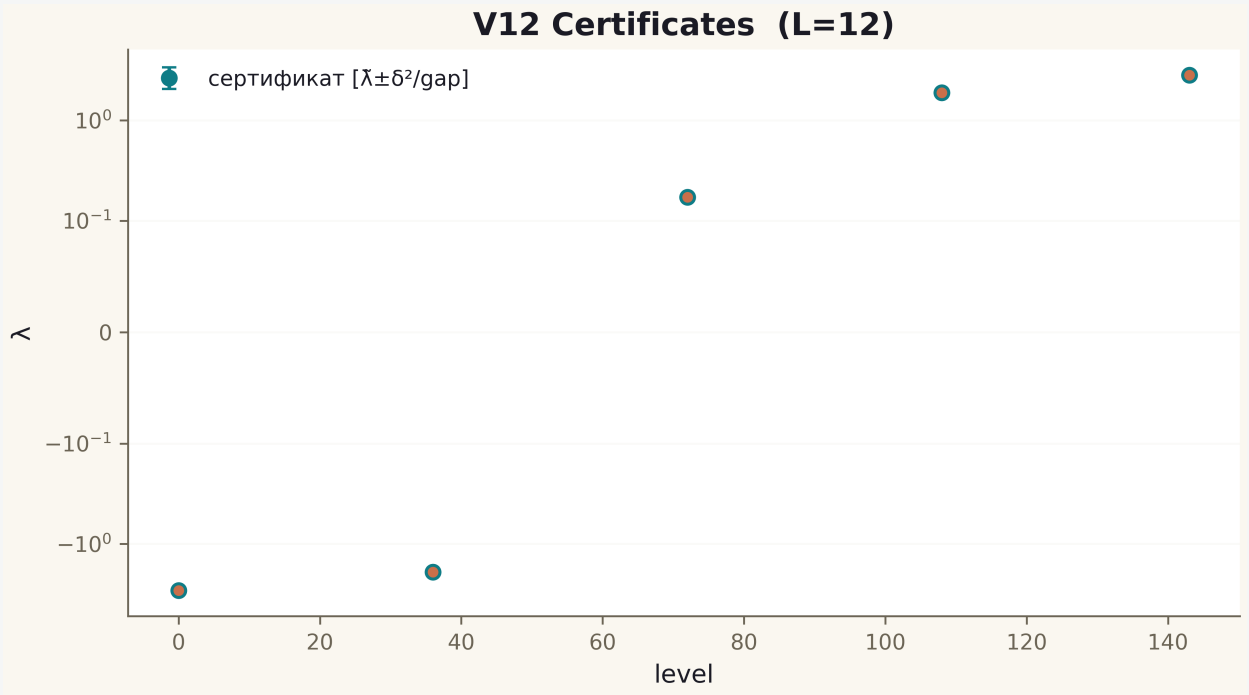


Fig. 1. Campaign V12 data (600 dpi, file figures_hires).

The key numerical results of the series now carry guaranteed intervals, not decimal tolerances. The certificate infrastructure transfers to any future experiment of the package.